

A. Given that 104 avalanches occurred during the past 10 years:

- i) what is the expected number of avalanches that should happen on any given day of the week if there is no association between day and probability of avalanche and
- ii) give a large sample 95% confidence interval for the number of avalanches seen on a given day of the week if there is no association between day and probability of avalanche

B. Based on 104 avalanches occurring overall, perform the appropriate two-sided large sample test of H_0 : that the probability of an avalanches is independent of whether or not people are skiing on that day (remember that people are skiing on Saturdays & Sundays only) at $\alpha = 0.01$